Advanced Programming for Robotics

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

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Advanced Programming for Robotics (A11883)

Short Title:	Advanced Programming Robotics		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Software and Web Development		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

Advanced Programming Concepts module teaches the learner further programming concepts and techniques including strings, recursion, pointers, structures and vectors. Further Object Oriented concepts are also introduced in this module.

Programmes

ADPC-0001	BEng (Hons) in Automation Engineering with Data Intelligence (`WD_EAUTO_B`)	2 / 3 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	2 / 3 / M
	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	2 / 3 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	2 / 3 / M

Indicative Content

- Strings
- Pointers
- Recursion
- Structures
- Vectors

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Apply problem solving strategies to various computing problems of increasing complexity.
2. Analyse different programming concepts and apply a particular programming problem.
3. Design, develop, test and document applications using advanced programming constructs and data structures.
4. Develop applications to control sensor inputs and outputs connected to a microcontroller.

Learning and Teaching Methods

- This course will be presented by a combination of lectures and computer-based practicals whilst capitalising on a web-enhanced learning environment
- The lectures will be used to introduce new topics and their related concepts
- The emphasis on course delivery will be hands-on, problem solving both individually and in small class groups using problem based worksheets during the practical sessions
- Self-directed learning will be encouraged throughout the duration of the module

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
In-Class Assessment	20%	1,2,3
Practical	80%	1,2,3,4

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Deitel, P. and H. Deitel. _C++ How to Program (Early Objects Version)_. New Jersey: Pearson Education, 2014.
- Savitch, W. _Problem Solving with C++_. 9th. New Jersey: Pearson Education, 2015.

Requested Resources

- Room Type: Computer Lab